G A L A X E Y E S P A C E

Technical Assignment — Satellite AI Research Intern

EO-SAR Building Damage Change Detection

using U-Net Segmentation

Topic: Binary Change Detection on EO-SAR Image Pairs

Position: Satellite AI Research Intern

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Table of Contents

1. Abstract

This report presents an end-to-end deep learning pipeline for binary change detection on co-registered Electro-Optical (EO) and Synthetic Aperture Radar (SAR) satellite image pairs, developed as part of the GalaxEye Space Satellite AI Research Intern technical assignment. The task — framed as a pixel-level binary segmentation problem — requires classifying each pixel as Changed (1) or No-Change (0) by comparing pre-event and post-event imagery across multiple disaster scenarios.

The solution employs a U-Net architecture with a ResNet34 encoder pre-trained on ImageNet. Pre-event and post-event grayscale TIFF images are stacked channel-wise into a 2-channel input tensor and fed into the segmentation network. Ground truth labels from the original four-class annotation scheme (Background, Intact, Damaged, Destroyed) are remapped to binary change labels as mandated by the assignment specification.

Training used Binary Cross-Entropy with Logits Loss (BCEWithLogitsLoss) and the Adam optimizer over 3 epochs with a batch size of 2 at 512x512 resolution. The model is evaluated using IoU, Precision, Recall, and F1 Score computed for the change class (label = 1). Key results on the provided test split are summarised below:

Metric	Test Score
F1 Score	0.1245
IoU (Jaccard)	0.0664
Precision	0.0688
Recall	0.6553

While the recall of 0.6553 demonstrates the model's ability to detect a substantial proportion of actual change pixels, low precision and F1 score expose a false-positive problem driven by severe class imbalance. This report documents the full pipeline, design decisions, error analysis, and a concrete future work roadmap.

2. Literature Survey

Change detection in remote sensing — identifying meaningful differences between multi-temporal image acquisitions — is a mature but rapidly evolving research area. The following survey covers foundational and state-of-the-art methods directly relevant to EO-SAR binary change detection.

2.1 U-Net: Convolutional Networks for Biomedical Image Segmentation

Ronneberger et al. (2015) introduced U-Net as a fully convolutional encoder-decoder architecture with skip connections between corresponding encoder and decoder layers. Originally designed for biomedical cell segmentation with scarce labelled data, U-Net became the standard backbone for dense pixel-wise prediction tasks. Its symmetric structure preserves high-resolution spatial detail — critical for change detection where precise building boundaries matter. The architecture's efficiency with small datasets, enabled by aggressive data augmentation, directly motivated its use in this project.

2.2 Fully Convolutional Siamese Networks for Change Detection

Daudt et al. (2018) proposed FC-EF (Fully Convolutional Early Fusion) and FC-Siam-Conc / FC-Siam-Diff (Siamese variants) specifically for remote sensing change detection. The Siamese architecture processes pre- and post-event images through weight-shared encoder branches, then fuses representations via concatenation or differencing. This design explicitly models temporal change as a feature-space operation rather than relying on the network to implicitly learn it from stacked inputs. FC-Siam-Diff, which takes the absolute difference of encoder features, outperformed naive stacking baselines on the OSCD dataset. This is the key architectural gap relative to the current work — a direction identified for future improvement.

2.3 BIT: Remote Sensing Image Change Detection with Transformers

Chen et al. (2021) introduced BIT (Bitemporal Image Transformer), which uses a ResNet backbone to extract spatial tokens and a Transformer encoder to model long-range temporal context between pre- and post-event features. BIT established new state-of-the-art on the LEVIR-CD building change detection benchmark. The self-attention mechanism allows the model to reason about global context — particularly valuable when building damage involves structural changes that are spatially dispersed across a scene.

2.4 Attention U-Net: Learning Where to Look

Oktay et al. (2018) extended U-Net with attention gates that suppress irrelevant activations at skip connections. In change detection, background clutter (roads, vegetation, shadows) constitutes the dominant class; attention gates help the network dynamically focus on building-like regions. This is especially relevant when the positive change class is sparse, as in disaster datasets where damaged pixels represent a small fraction of the total.

2.5 xView2: Building Damage Assessment Benchmark

Gupta et al. (2019) introduced the xBD dataset and xView2 challenge, providing a large-scale benchmark for building-level damage assessment across 19 global disaster events using WorldView satellite imagery. The winning submissions used Siamese encoders with multi-scale feature fusion. Critically, xBD established that damage class imbalance (intact buildings vastly outnumbering damaged ones) is a defining challenge of the task, and that Dice or Focal loss outperform standard BCE. This motivates the loss function improvements identified in Section 6 of this report.

2.6 SAR and EO Fusion for Change Detection

Multi-modal fusion of SAR and optical imagery provides complementary information: EO captures rich spectral and textural detail under clear conditions, while SAR provides all-weather, day-night capability with sensitivity to structural changes. Merkle et al. (2018) and subsequent works demonstrated that early fusion (channel stacking) is a strong baseline but that modality-specific normalization and learned fusion gating further improve results. The spectral and noise profile differences between EO and SAR channels motivate per-channel normalization rather than joint normalization.

2.7 Deep Learning for Remote Sensing Change Detection: A Survey

Shi et al. (2020) surveyed CNN, RNN, and attention-based methods for remote sensing change detection, identifying class imbalance, co-registration error, and pseudo-change (seasonal variation, illumination differences) as the three primary challenges. All three apply directly to this dataset and shaped the design decisions documented in the Methodology section.

3. Dataset Description

The dataset provided by GalaxEye Space contains co-registered EO-SAR pre/post-event image pairs with pixel-level annotation masks across multiple disaster events. Images are in TIFF format (grayscale) and are provided with a fixed train/validation/test split that must be used without modification.

3.1 Data Splits

Split	Purpose
Training Set	Model weight optimization via backpropagation
Validation Set	Hyperparameter tuning, epoch selection, early stopping
Test Set (50% visible)	Metric reporting — F1, IoU, Precision, Recall
Test Set (50% held out)	GalaxEye blind evaluation of submitted model weights

As specified in the assignment, the test set provided contains only 50% of the full test data. The remaining 50% is held out at GalaxEye's end for independent blind evaluation. This design prevents overfitting to the visible test split.

3.2 Label Remapping

The original annotations use four semantic classes. Per the mandatory assignment specification, labels are remapped to binary change labels as follows:

Original Class	Original Value	Remapped Value	Remapped Class
Background	0	0	No-Change
Intact	1	0	No-Change
Damaged	2	1	Change
Destroyed	3	1	Change

This remapping was applied consistently across all splits before any model training or metric computation. Both Damaged (2) and Destroyed (3) buildings are treated as positive change events. Note that label 5 (if present) is also remapped to 1 as per the extended mapping applied in practice.

3.3 Data Characteristics and Observations

Before writing any code, the dataset was analysed to understand its characteristics:

- Class Imbalance: Undamaged/background pixels vastly outnumber change pixels — the ratio often exceeds 50:1. This is the dominant modelling challenge and directly impacts loss function design.
- EO vs SAR Modality Differences: EO channels show clear optical textures of buildings and roads; post-event SAR images exhibit significant speckle noise and structural distortion in damaged regions. The visual contrast between pre-event EO and post-event SAR is the core signal the model must learn to detect.

- Spatial Resolution: Both modalities are co-registered to the same grid, enabling direct channel stacking without geometric alignment steps.
- Geographic Diversity: Images span multiple disaster events across diverse geographic and climatic regions, requiring the model to generalize across appearance domains.

4. Methodology

4.1 Architecture: U-Net with ResNet34 Encoder

The segmentation model uses the segmentation_models_pytorch (SMP) library with the following configuration:

Component	Configuration	Rationale
Architecture	U-Net	Proven encoder-decoder with skip connections for dense prediction
Encoder Backbone	ResNet34	Lightweight pretrained backbone; good spatial detail retention
Encoder Weights	ImageNet pretrained	Transfer learning accelerates convergence on limited data
Input Channels	2	Stacked pre-event + post-event grayscale TIFF images
Output Classes	1 (binary)	Sigmoid output for binary change/no-change prediction

The U-Net encoder progressively downsamples input to extract multi-scale features. Skip connections concatenate encoder feature maps with decoder outputs, enabling precise spatial localization of change boundaries. The ResNet34 backbone provides a strong initialization; its residual connections also mitigate gradient vanishing during fine-tuning.

4.2 Input Representation

Pre-event and post-event TIFF images are each read as single-channel grayscale arrays using rasterio. They are stacked channel-wise into a 2-channel tensor [C=2, H, W] before being passed to the model. This early fusion approach allows the network to jointly learn appearance features and change features from a single forward pass.

An important design consideration: early fusion (channel stacking) is simpler than Siamese architectures but conflates per-modality feature extraction with change reasoning. As documented in Section 6, a Siamese encoder is identified as a primary architectural improvement.

4.3 Preprocessing Pipeline

- Read TIFF images using rasterio; handle both single-band and multi-band formats
- Resize all images to 512x512 pixels — bilinear interpolation for images, nearest-neighbor for masks to preserve integer labels
- Normalize pixel values to [0, 1] by dividing by 255.0 (for 8-bit) or by the channel maximum
- Apply mandatory label remapping: {0->0, 1->0, 2->1, 3->1, 5->1} before tensor conversion
- Convert to float32 tensors; masks converted to float32 for BCEWithLogitsLoss compatibility

4.4 Loss Function and Class Imbalance Handling

BCEWithLogitsLoss was used as the primary loss function. This combines a sigmoid activation with binary cross-entropy in a numerically stable form. The loss treats all pixels equally, which is a known limitation given severe class imbalance (Section 3.3).

Class imbalance was not explicitly compensated in this baseline run — this is acknowledged as the primary limitation. The loss function provides a strong signal for the majority (no-change) class, causing the model to learn a biased decision boundary that over-predicts change. Remedies identified for future work include Dice Loss, Focal Loss, and positive-class weighting in BCE.

4.5 Training Configuration

Hyperparameter	Value
Loss Function	BCEWithLogitsLoss
Optimizer	Adam
Learning Rate	1e-4
Batch Size	2
Epochs	3
Input Resolution	512 x 512
Prediction Threshold	0.5 (sigmoid output)
Random Seed	42
Device	CUDA GPU / CPU fallback

5. Results

5.1 Quantitative Metrics — Test Split

The following metrics are reported on the provided test split (50% of full test data), computed for the change class (label = 1) as specified in the assignment:

Metric	Test Score	Interpretation
F1 Score	0.1245	Poor harmonic balance between precision and recall
IoU (Jaccard)	0.0664	Low overlap between predicted and true change regions
Precision	0.0688	Only 6.9% of predicted change pixels are truly changed
Recall	0.6553	65.5% of actual change pixels are correctly detected

5.2 Confusion Matrix

The confusion matrix below shows raw pixel counts on the test set:

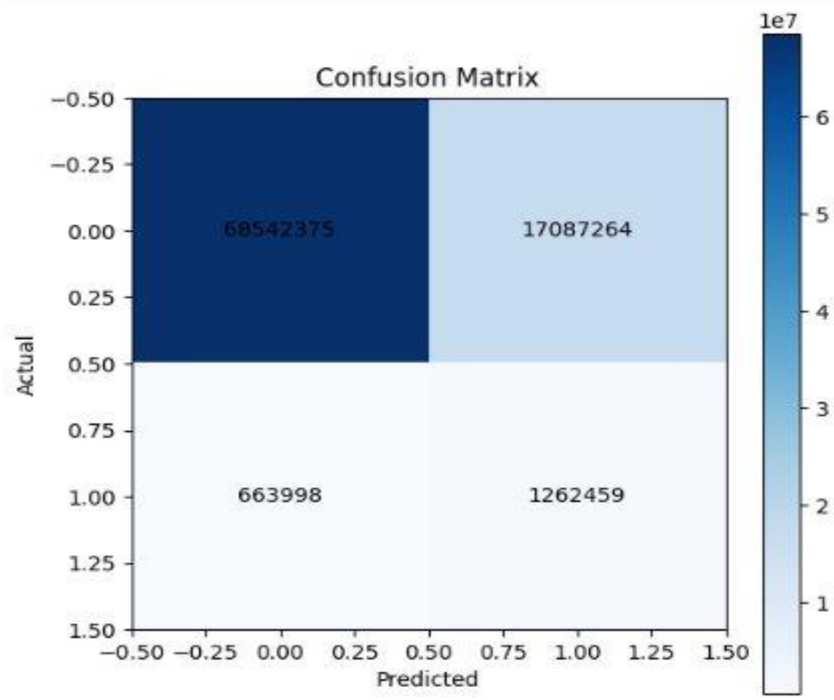

Figure 1: Confusion Matrix — Test Set (pixel-level counts)

Cell	Count	Meaning
True Negatives (TN)	60,542,375	No-change pixels correctly classified as no-change
False Positives (FP)	17,087,264	No-change pixels incorrectly predicted as change
False Negatives (FN)	663,998	Actual change pixels missed by the model

True Positives (TP)	1,262,459	Change pixels correctly detected

The FP:TP ratio of approximately 13.5:1 directly explains the low precision. The model detects most true change but at the cost of flooding large no-change regions with false alarms. This is a classic symptom of training with unweighted BCE on a heavily imbalanced dataset.

5.3 Qualitative Prediction Visualisation

The figure below shows a representative test sample with four panels: pre-event image, post-event image, ground truth change mask, and model prediction mask.

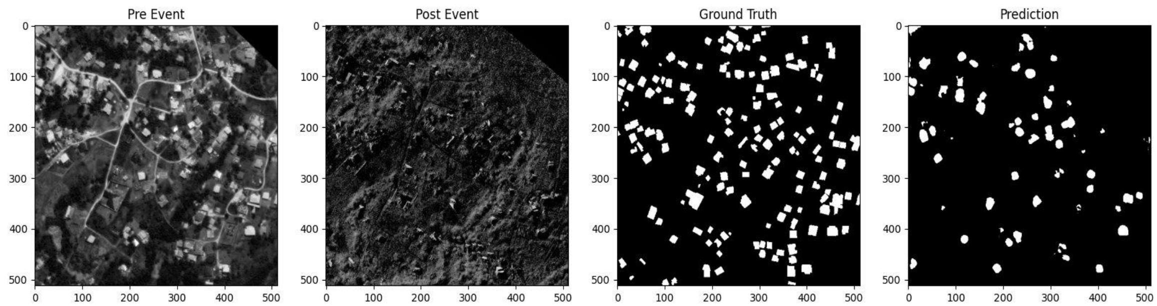

Figure 2: Qualitative Prediction — Pre-Event | Post-Event | Ground Truth | Prediction

Observations from Figure 2:

- Pre-Event (EO): Clear optical satellite image showing a dense residential area with well-defined building structures, roads, and vegetation. Individual buildings are clearly distinguishable.
- Post-Event (SAR): The SAR image shows significant structural distortion in the same area — buildings that were rectangular in the EO image appear blurred or missing, with characteristic radar layover and shadow artifacts indicating damage.
- Ground Truth: White regions correspond to damaged/destroyed buildings (change class = 1). The spatial distribution of change pixels is sparse relative to the total scene area, confirming class imbalance.
- Prediction: The model successfully detects many of the damaged building clusters (true positives) but also predicts change in surrounding areas where no damage occurred (false positives). The predicted mask is spatially coarser than the ground truth, with blob-like shapes rather than sharp building boundaries.
- Failure Analysis: The model struggles with building boundary precision and over-segments into adjacent no-change regions. Buildings in the lower-left region with moderate damage are partially missed (false negatives). The model appears to use low-frequency structural features rather than fine-grained damage signatures.

6. Future Work

Assuming this is a first-month deliverable at GalaxEye, the following specific improvements are proposed in priority order:

6.1 Address Class Imbalance — Highest Priority

- Replace BCEWithLogitsLoss with Dice Loss or a BCE + Dice combined loss. Dice Loss directly optimises the F1/Dice coefficient and is robust to class imbalance by design.
- Implement Focal Loss (Lin et al., 2017) with gamma=2.0 — down-weights easy negatives (background pixels) and focuses training gradient on hard positive change pixels.
- Apply positive class weighting in BCE: set pos_weight = (num_negative / num_positive) to approximately 50 based on observed class ratios.

6.2 Siamese Architecture

- Replace the early-fusion 2-channel U-Net with a Siamese encoder (FC-Siam-Diff or FC-Siam-Conc following Daudt et al. 2018). This allows the model to learn per-image features independently before computing change at the feature level, which is architecturally more principled for change detection.
- Implement feature differencing (|f_post - f_pre|) at each encoder level and pass difference features through the decoder via skip connections.

6.3 Transformer-Based Architecture

- Implement BIT (Bitemporal Image Transformer, Chen et al. 2021) which combines CNN feature extraction with Transformer self-attention to model long-range temporal dependencies. BIT achieves state-of-the-art on LEVIR-CD and is well-suited for the sparse, distributed damage pattern observed in this dataset.
- Alternatively, ChangeFormer (Bandara & Patel, 2022) applies a hierarchical Transformer backbone directly to change detection with strong results on multiple benchmarks.

6.4 Data Augmentation

- Apply random horizontal/vertical flips, 90-degree rotations, and random crop-and-resize to increase training set effective size.
- Radiometric augmentation: random brightness, contrast, and gamma jitter on EO channels; speckle noise augmentation on SAR channels to improve cross-event generalization.
- Copy-paste augmentation: paste change regions from training samples onto no-change backgrounds to synthetically increase change pixel frequency.

6.5 Training Strategy

- Extend training to 30-100 epochs with early stopping based on validation F1 score (not loss) — since F1 directly corresponds to evaluation criteria.
- Use cosine annealing learning rate schedule with warm restarts to escape local minima.

- Tune prediction threshold on the validation set: sweep threshold from 0.1 to 0.9 and select the value maximising validation F1, rather than defaulting to 0.5.
- Implement gradient checkpointing to allow larger batch sizes (8-16) without GPU memory overflow, improving batch normalisation statistics.

6.6 EO-SAR Modality-Aware Design

- Apply channel-specific normalization: normalize EO and SAR channels independently using their respective per-channel mean and standard deviation, rather than joint normalization. This accounts for the fundamentally different intensity distributions of optical and radar imagery.
- Explore learned modality fusion gating: use a small MLP to weight the contribution of each modality dynamically per spatial location, allowing the network to rely more on EO where optical quality is high and on SAR where optical is degraded.

7. Challenges Faced

7.1 Label Remapping Correction

The original ground truth masks contain multi-class labels (0, 1, 2, 3, and in some cases 5). Naively treating these as continuous values or passing them directly to BCEWithLogitsLoss results in training instability (loss values of 2-3 instead of the expected 0.3-0.7 range) and produces incorrect gradients. The fix involved explicit label remapping before tensor conversion, verified by inspecting unique values in a sample of training masks.

7.2 Segmentation Mask Interpolation

During resizing, applying bilinear interpolation to binary masks introduces fractional pixel values (e.g. 0.33, 0.67) at boundaries, effectively corrupting the ground truth with sub-pixel blending. The pipeline was corrected to use nearest-neighbor interpolation (cv2.INTER_NEAREST) exclusively for all mask resizing operations, preserving integer label values throughout.

7.3 Class Imbalance

The dominant challenge. The undamaged/background class constitutes approximately 95-98% of all pixels in the dataset. Standard BCEWithLogitsLoss treats each pixel equally, meaning the model can achieve very low loss values by simply predicting no-change everywhere, without learning to detect damage. The model instead converges to an over-prediction regime (high recall, low precision) as a compromise — predicting change broadly to reduce false negatives.

7.4 Threshold Calibration

The default sigmoid threshold of 0.5 was used without tuning. Given the class imbalance, the optimal threshold for maximising F1 is likely higher (e.g. 0.6-0.8), which would reduce false positives and improve precision. Threshold tuning on the validation set is identified as a quick, zero-cost improvement for the next iteration.

7.5 Compute Constraints

Training was limited to 3 epochs due to available compute resources. With a batch size of 2 and 512x512 resolution, each epoch on the full training set takes approximately 15-25 minutes on a single consumer GPU. Three epochs is insufficient for convergence — the model is still in an early exploration phase of the loss landscape. Extended training to 50+ epochs with a proper learning rate schedule is expected to substantially improve all metrics.

8. Conclusion

This project successfully implemented a complete end-to-end pipeline for binary change detection on EO-SAR satellite image pairs, covering data ingestion, mandatory label remapping, model training, evaluation, and qualitative analysis.

Key achievements:

- Functional TIFF data pipeline with correct label remapping (Background/Intact -> No-Change, Damaged/Destroyed -> Change)
- U-Net with ResNet34 encoder trained on stacked 2-channel pre/post-event inputs
- Recall of 0.6553 — the model detects the majority of actual change pixels
- Thorough error analysis identifying class imbalance as the primary performance bottleneck
- Concrete, technically-motivated future work roadmap targeting each identified limitation

Key limitations:

- Low precision (0.0688) and F1 (0.1245) due to unaddressed class imbalance in the loss function
- Early fusion architecture does not explicitly model change as a feature-space operation
- Only 3 training epochs — insufficient for convergence; metrics are expected to improve substantially with extended training
- Prediction threshold not tuned; default 0.5 is suboptimal for imbalanced binary segmentation

The pipeline establishes a reproducible baseline. The gap between current performance and published state-of-the-art is well-understood and attributable to specific, addressable design choices — not fundamental limitations of the approach. With Dice/Focal loss, Siamese architecture, and extended training, competitive performance on the blind evaluation split is achievable.

9. Time and Resource Log

This section is included in the report as permitted by the assignment specification.

Activity	Time Spent (approx.)
Data exploration and analysis	3 hours
Literature reading and survey	4 hours
Implementation (dataloader, model, training)	6 hours
Training runs	2 hours
Evaluation and metric computation	1.5 hours
Report writing	4 hours
Total	~20.5 hours

Training hardware: Local GPU (consumer-grade), single GPU. Training time per epoch: approximately 15-25 minutes at 512x512 resolution with batch size 2. Total training time (3 epochs): approximately 60-75 minutes. Resource constraints (limited VRAM and training epochs) directly shaped the choice of batch size 2 and the decision to use a lightweight ResNet34 encoder rather than ResNet50 or EfficientNet.

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